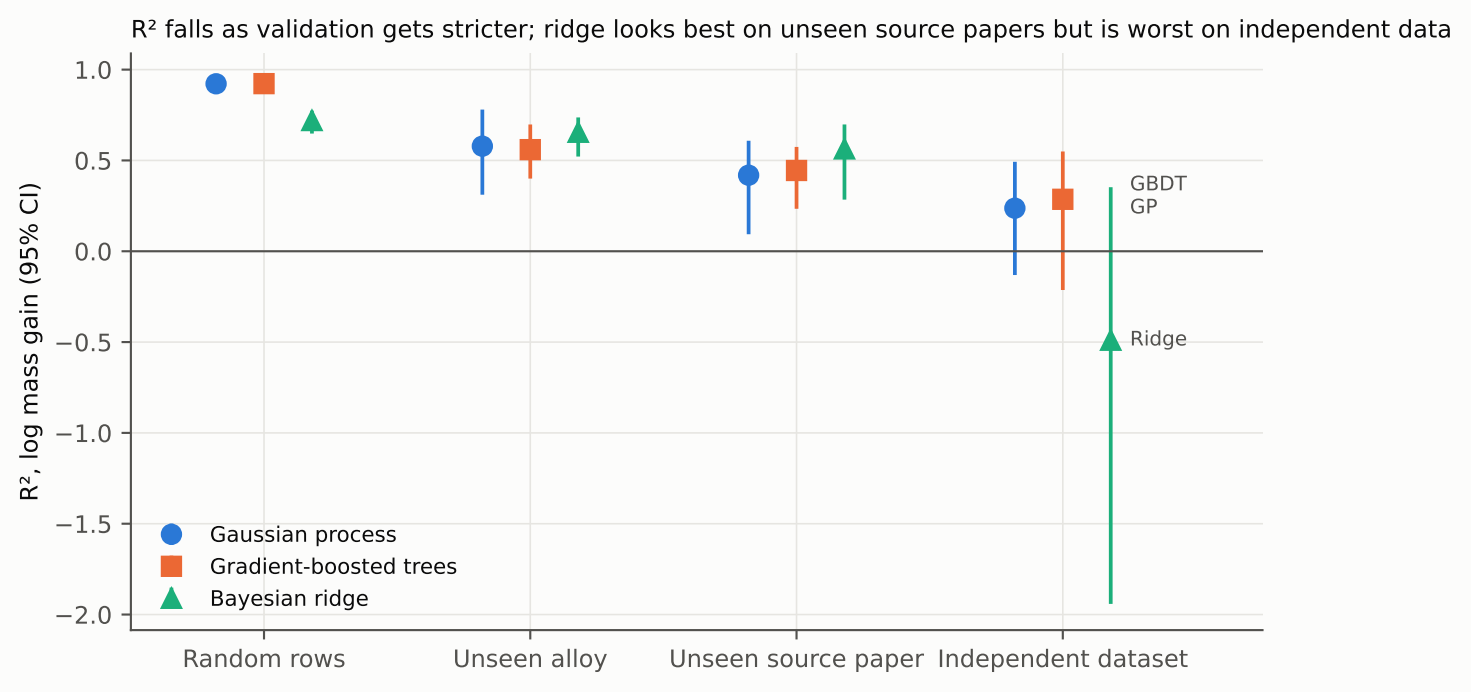
R² falls as validation gets stricter; ridge looks best on unseen source papers but is worst on independent data
R², log mass gain (95% CI)
1.0
0.5
0.0
−0.5
−1.0
−1.5
−2.0
GBDT
GP
Ridge
Gaussian process
Gradient-boosted trees
Bayesian ridge
Random rows
Unseen alloy
Unseen source paper
Independent dataset